

Assignment No	23
Title	Forms in React js
Objective	Form validation in react js
Roll No	MCA2511

Code:

[Form1.js](#)

```
import React,{useState} from "react";
import '../App.css';
import { data } from "react-router-dom";

function Form1(){
  const [formData, setFormData] = useState({
    username:'',
    email:'',
    password:'',
    confirmPassword:'',
  });

  const [errors, setErrors] = useState({});
  const handleChange = (e) => {
    const { name , value } = e.target;
    setFormData({
      ...formData,
      [name]:value,
    });
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    const newErrors = validateForm(formData);
    setErrors(newErrors);
    if(Object.keys(newErrors).length === 0){
      alert('Form submitted successfully !!');
    }
  };
}
```

```
    }
    else{
        alert('Submission failed due to errors !!');
    }
};

const validateForm = (data) =>{
    const errors = {};
    if(!data.username.trim()){
        errors.username = 'Username is required !!';
    }
    else if(data.username.length < 6)
    {
        errors.username = 'Username must be at least 6 characters';
    }
    if(!data.email.trim())
    {
        errors.email = 'Email is required !!';
    }

    if(!data.password.trim())
    {
        errors.password = 'Password is required';
    }
    else if(data.password.length < 8){
        errors.password = 'password must be atleast 8 character
long';
    }
    if(data.confirmPassword != data.password){
        errors.confirmPassword = 'Password does not match !!';
    }
    return errors;
}

return (
    <>
    <form onSubmit={handleSubmit} className="form">
        <h1>Registration Form</h1>
        <div>
            <label>Username</label><br></br>
            <input
```

```
        type="text"
        name="username"
        value={formData.username}
        onChange={handleChange}
    />
    {errors.username && <p className="error">{errors.username}</p>}
</div><br></br>

<div>
    <label>Email</label><br></br>
    <input
        type="email"
        name="email"
        value={formData.email}
        onChange={handleChange}
    />
    {errors.email && <p className="error">{errors.email}</p>}
</div><br></br>

<div>
    <label>Password</label><br></br>
    <input
        type="password"
        name="password"
        value={formData.password}
        onChange={handleChange}
    />
    {errors.password && <p className="error">{errors.password}</p>}
</div><br></br>

<div>
    <label>Confirm Password</label><br></br>
    <input
        type="password"
        name="confirmPassword"
        value={formData.confirmPassword}
        onChange={handleChange}
    />
    {errors.confirmPassword && (
        <p className="error">{errors.confirmPassword}</p>
    )}
```

```
        </div><br></br>

        <button type="submit">Submit</button>
    </form>
</>
);
}
export default Form1;
```

App.css

```
.form {
  width: max-content;
  background-color: rgb(155, 255, 188);
  font-family: 'Courier New', Courier, monospace;
  border-radius: 20px;
  padding: 50px;
  text-align: center;
  margin: 40px 450px;
}

.error {
  color: red;
  margin: 0;
}
```

[index.js](#)

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
import reportWebVitals from './reportWebVitals';
import './App.css';
import Form1 from './Components/Form1';

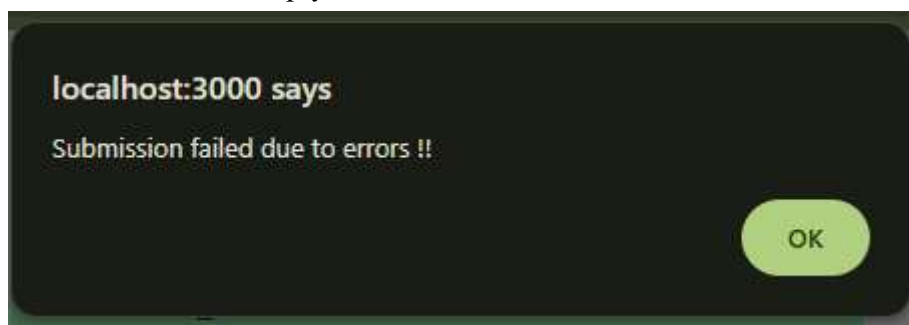
const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <Form1/>
  </React.StrictMode>
);
reportWebVitals();
```

Output:



A registration form titled "Registration Form" on a light green background. The form contains four input fields labeled "Username", "Email", "Password", and "Confirm Password", each with a corresponding text input box. Below the input fields is a "Submit" button.

Error if submitted Empty



Registration Form

Username

Username is required !!

Email

Email is required !!

Password

Password is required

Confirm Password

Submit

Validation on Username

Registration Form

Username

Username must be at least 6 characters

Validation on Email



Email

userxamplecom

Please include an '@' in the email address. 'userxamplecom' is missing an '@'.

This image shows a web form with a light green background. At the top, the label "Email" is centered. Below it is a text input field containing the text "userxamplecom". A white error message box with a yellow border and a yellow exclamation mark icon is positioned below the input field. The message reads: "Please include an '@' in the email address. 'userxamplecom' is missing an '@'."

Validation on Password



Password

...

password must be atleast 8 character long

Confirm Password

...

Password does not match !!

Submit

This image shows a web form with a light green background. It has two password fields. The first field is labeled "Password" and contains three dots. Below it, a red error message says "password must be atleast 8 character long". The second field is labeled "Confirm Password" and also contains three dots. Below it, a red error message says "Password does not match !!". At the bottom center is a "Submit" button.

Alert if Validation Successful and Form submitted Successfully



localhost:3000 says

Form submitted successfully !!

OK

This image shows a dark-themed alert box. It contains the text "localhost:3000 says" and "Form submitted successfully !!". At the bottom right is a green "OK" button.